PROJECT REPORT

Project Background

Big Box Warehouses (BBW) is in the process of modernizing its stores. One critical aspect BBW must improve in its stores is the checkout process. One of the main customers' complaints the company receives is the long lines at the checkouts, with customers frequently reporting waiting times in checkout queues of 30 minutes and above during the busier periods of the day.

BBW has decided to replace its cashier checkout stations with more modern and efficient ones and is introducing self-checkout stations as well. CAU Corporation checkout stations models were selected for both cashier and self-checkout stations. This project covers the implementation of the new checkout stations in BBW's stores.

After a proof-of-concept implementation in store #011, the company has selected store #115 as the next store to install the new checkout stations. Store #115 currently has 10 cashier checkout lanes.

Data Analysis

To perform a discrete event simulation of the BBW's checkout process we will need distributions to generate the following:

- Inter-arrival times for customers
- Shopping Time (excluding checkout and exit)
- Checkout processing times

The data was analyzed using a goodness of fit software to find the probability distribution that best fits the data.

Inter-arrival Times

The store provided the total customers arrivals to the store by hour. Plotting the arrivals per hour of day for each day (Figure 1), we can observe the following:

- Arrivals for the weekend (Saturday and Sunday) are higher than arrivals for weekdays (Monday to Friday).
- Arrivals vary throughout the day.

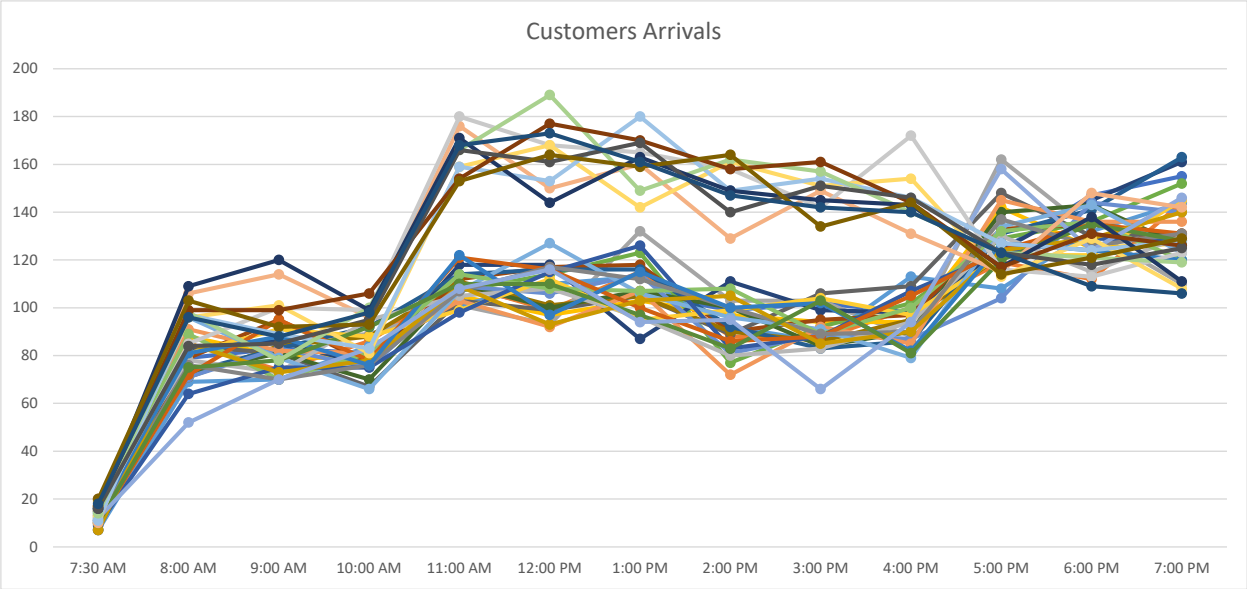


Figure 1: Customers' Arrivals Plot

Since the cashiers work in shifts of 3.00 hours, and the graph indicates the arrivals variation during the day follows the same schedule, the following arrivals rates were calculated:

Table 1: Arrival Rates per Shift Type

Shift	Arrival Rate (customers/ hour)	Poisson Distribution?
Weekday 7:30 AM – 8:00 AM	27.00	Yes, Chi-Squared p-value = 0.66
Weekday 8:00 AM – 11:00 AM	80.00	Yes, Chi-Squared p-value = 0.82
Weekday 11:00 AM – 2:00 PM	108.00	Yes, Chi-Squared p-value = 0.86
Weekday 2:00 PM – 5:00 PM	93.00	Yes, Chi-Squared p-value = 0.74
Weekday 5:00 PM – 8:00 PM	133.00	Yes, Chi-Squared p-value = 0.66
Weekend 7:30 AM – 8:00 AM	30.00	Yes, Chi-Squared p-value = 0.41
Weekend 8:00 AM – 11:00 AM	96.00	Yes, Chi-Squared p-value = 0.16
Weekend 11:00 AM – 2:00 PM	164.00	Yes, Chi-Squared p-value = 0.98
Weekend 2:00 PM – 5:00 PM	149.00	Yes, Chi-Squared p-value = 0.64
Weekend 5:00 PM – 8:00 PM	122.00	Yes, Chi-Squared p-value = 0.82

If the customers arrival rates follow a Poisson Distribution, the interarrival times follow an Exponential Distribution. Therefore, the goodness of fit test was performed to confirm that we can use the Poisson distribution to generate the arrivals.

Checkout Processing Times

The following distributions were selected for the checkout times data provided.

Table 2: Checkout Processing Times Distributions

Data	Distribution	Parameters	Mean (minutes)	Test p-value (*)
Current Cashiers Stations	LogNormal Distribution	Ln mu = 1.83 Ln sigma = 0.48	7.00	KS (0.88) AD (0.87)
CAU Cashiers Stations	LogNormal Distribution	Ln mu = 1.51 Ln sigma = 0.16	4.59	KS (0.98) AD (0.99)
CAU Self-Checkout Stations	LogNormal Distribution	Ln mu = 2.03 Ln sigma = 0.15	7.70	KS (0.69) AD (0.82)

(*) KS => Kolmogorov-Smirnov, AD => Anderson-Darling

Shopping Time

The time customers spend shopping can be calculated by generating the number of items purchased by customers and using the shopping time per item distribution provided in the data. The goodness of fit analysis indicates that the items purchased per customer follow a Negative Binomial distribution with parameters k = 7 and p = 0.25, and a mean of 21.00 items per customer. The Chi-Square p-value for the test is 0.94. The data provided for this project indicates that for store #115, the shopping time per item purchased follows a uniform distribution with minimum of 30.00 seconds and maximum of 90.00 seconds (mean 60.00 seconds per item).

System Conceptual Models

The design alternatives for the checkout area fall into two categories based on the self-checkout queue option. The single queue alternatives, illustrated in Figure 2, have N cashier checkout lanes, each with its own queue; and M self-checkout lanes, each lane with two stations, but only one queue for the M lanes. The multiple queue alternatives, illustrated in Figure 3, have N cashier checkout lanes and M self-checkout lanes, each lane with its own queue; each self-checkout lane has two stations.

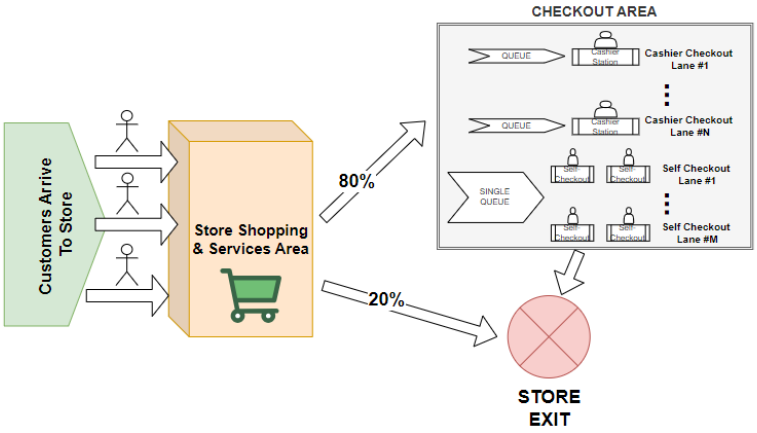


Figure 2: Conceptual Model -Single Queue for Self-Checkout Lanes

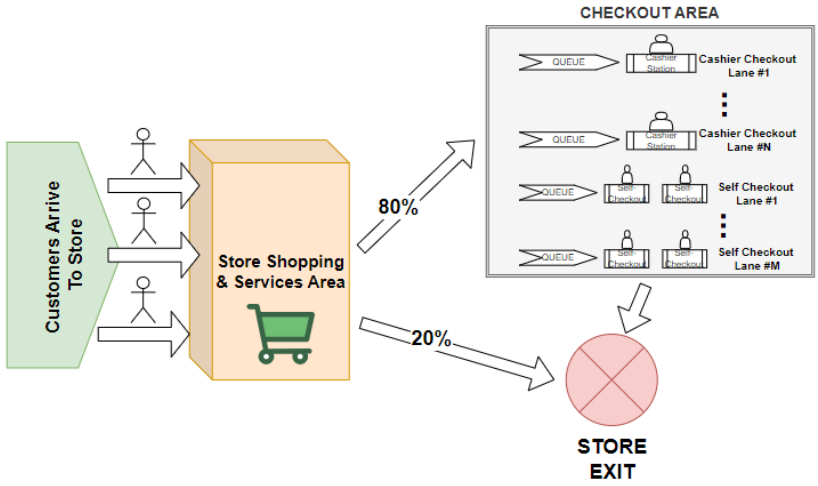


Figure 3: Conceptual Model - Multiple Queues for Self-Checkout Lanes

Store #115 currently has 10 checkout lanes, but the new stations are narrower and allow up to 25.00% more lanes in the same space. Therefore, the limit for the N + M is 12 lanes. Since one cashier can attend up to 6 self-checkout lanes (12 stations) and not all the 12 lanes can be self-checkout, then it would be inefficient to have more than 6 self-checkout lanes.

Discrete Event Simulation Study

The logic used to build the simulation model is shown in the report's Appendix 1.

Model Validation

Although no validation data was available, some tests were performed to check the simulation model was behaving adequately. Here are some of the tests and results:

- A simulation run, with replication length of 1 week and 30 replications, was performed utilizing the current cashiers processing times and 10 cashiers lanes to simulate current conditions. The average waiting time was 27.61 minutes, with the last two shifts on weekend days exceeding the hour and 20.00 minutes waiting. The average checkout station utilization was 0.82 but reached 1.00 on the weekends.
- A Simulation was run with arrival rates to be 25.00% of the actual rates. The average waiting time dropped to zero, and average utilization dropped to 0.23.
- A Simulation was run with arrival rates 75.00% higher than actual rates. The average waiting time jumped over 2.00 hours, and utilization increased close to 1.00 in 4 out of the 8 shifts.

It would have been ideal to have a set of output data to validate the model. Nonetheless, these validation tests, although they are not sufficient, showed the model is behaving as expected; and the "as-is" run showed very long wait times at peak hours, validating customers' complaints.

Checkout Area Design

The checkout area configuration should be able to handle the highest demand, so this period should be the one used in the study to determine the recommended configuration. According to the store information, the weekend foot traffic is higher than the weekdays, and the store experiences a 15.00% increase in foot traffic during the holidays. Hence the arrival rates used correspond to a weekend date, increased by 15.00%.

All the dates simulated were after the study date of May 29, 2026. A total of 50 replications, of 1.00 days in length, were run for each of the scenarios in Table 3. The checkout processing times for the new stations were used for all the scenarios. The table contains the mean and standard deviation for the average waiting time in queue for the day, as well as the absolute error for the mean (using 95% confidence level). The table also contains the max mean average waiting time in queue among the shifts, the average checkout stations utilization for the day and the max average utilization among the shifts.

Main observations derived from the results:

- The current 10 Cashier checkout lanes are not enough to handle the peak demand the store receives during the holidays.
- Configurations of 11 checkout lanes seem to be enough to handle the regular weekend and the holidays demand. Only 10 lanes configurations that utilize self-checkout lanes can satisfy the waiting time constraint.
- Only configurations with at least 6 cashier checkout lanes satisfy the constraint.
- The single queue for self-checkout stations seems to reduce the average waiting time by an average of 12.49% (7.19%, 23.60%, 6.67%) in comparison to the individual queues per lane, given the same number of self-checkout and cashiers checkout stations. However, using the single queue doesn't have an impact on the utilization of the cashiers checkout stations.

Table 3: Simulation Results for Peak Demand

Scenario	Number Cashiers Required	Average Waiting Time in Queue				Average Checkout Stations Utilization			
		Mean	Std. Dev.	Absolute Error	Shift Max	Self Average	Self Max	Cashier Average	Cashier Max
10 Cashier Checkout Stations	10	20.53	5.10	1.41	31.51	N/A	N/A	0.81	1.00
11 Cashier Checkout Stations	11	4.53	2.68	0.74	8.42	N/A	N/A	0.72	0.98
12 Cashier Checkout Stations	12	1.66	0.98	0.27	2.62	N/A	N/A	0.67	0.90
5 Self, 5 Cashier, individual Queues	6	11.13	4.79	1.33	16.99	0.64	0.94	0.86	1.00
5 Self, 6 Cashier, individual Queues	7	2.67	0.85	0.24	3.85	0.53	0.82	0.82	0.98
4 Self, 7 Cashier, individual Queues	8	1.35	0.28	0.08	1.78	0.45	0.68	0.79	0.96
4 Self, 6 Cashier, Single queue for Self	7	2.56	0.67	0.19	3.68	0.50	0.52	0.82	0.99
4 Self, 5 Cashier, Single queue for Self	6	9.73	4.14	1.15	14.26	0.57	0.76	0.86	1.00
5 Self, 5 Cashier, Single queue for Self	6	10.33	4.71	1.30	15.31	0.53	0.73	0.85	1.00
5 Self, 6 Cashier, Single queue for Self	7	2.04	0.62	0.17	2.79	0.47	0.66	0.82	0.98
4 Self, 7 Cashier, Single queue for Self	8	1.26	0.18	0.05	1.67	0.45	0.67	0.78	0.96

Based on the results, the recommended configuration is 5 self-checkout lanes (10 stations) and 6 cashiers checkout lanes, with single line for self-checkout lanes. The alternative of 4 self / 7 cashiers checkout stations also satisfies the peak demand but

increases the number of cashiers required. The single queue for self-checkout stations lowers the waiting time, and has a minimal, but positive impact on the utilization of the cashiers checkout stations.

Recommended Staffing

The recommended configuration was used to determine the number of cashiers (employees) required per shift. The number of cashier checkout stations was subsequently reduced to find what level still produces acceptable results (Average waiting time below 10 minutes) for each shift. Results are summarized in Table 4. This study assumed that all the self-checkout stations will be available with one cashier always attending, and the open cashier checkouts lanes will be manned for the whole shift. The store regular demand was used.

Table 4: Staffing Required per Shift

Shift	Cashier Checkouts Open	Cashiers Hours
Weekday 8:00 AM – 11:00 AM	2 stations	9.00 hours
Weekday 11:00 AM – 2:00 PM	3 stations	12.00 hours
Weekday 2:00 PM – 5:00 PM	4 stations	15.00 hours
Weekday 5:00 PM – 8:00 PM	4 stations	15.00 hours
Weekend 8:00 AM – 11:00 AM	3 stations	12.00 hours
Weekend 11:00 AM – 2:00 PM	4 stations	15.00 hours
Weekend 2:00 PM – 5:00 PM	5 stations	18.00 hours
Weekend 5:00 PM – 8:00 PM	5 stations	18.00 hours

Conclusions and Recommendations

The simulation study corroborated that implementing the self-checkout and new cashier checkout stations will have a positive impact on customers' satisfaction by significantly reducing the time the customers wait in queue. The recommended configuration for the checkout area is to install 5 self-checkout lanes (10 stations), with a single queue, and 6 cashier lanes with individual queues. This configuration satisfies the 10 minutes maximum waiting time in queue, even during the peak demand on the holidays. Since the new checkout stations allow up to 12 lanes to be installed on store #115, it is recommended to leave the remaining space open in case of future increases in foot traffic. The staffing levels required a total of 51.00 cashier hours per day from Monday to Friday, and a total of 63.00 hours per day during the weekends.

Appendix 1: Simulation Logic

